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title: "26-Level Vacuum Density Ladder: $\rho_{\text{vac}}^{(n)}$ Hierarchy via Ramanujan Zeta Regularisation and SCm Phonon Eq

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26-Level Vacuum Density Ladder

Abstract

We derive a complete 26-level vacuum density hierarchy $\rho_{\text{vac}}^{(n)}$ for $n = 1 \dots 26$, corresponding to the 26 independent dimensional spheres of the UQFF compressed gravity framework. Each level is governed by:

$$\rho_{\text{vac}}^{(n)} = \rho_{\text{SCm}} \cdot S_{26}^{(3)} \cdot \delta_n$$

where $\delta_n = (2\pi)^{n/6}$ encodes the dimensional scaling factor, $\rho_{\text{SCm}} = \rho_{\text{vac},0} \cdot H_{\text{SCm}} \cdot (1 - \kappa)$ is the SCm-corrected cosmological vacuum density, and $S_{26}^{(3)} = \sum_{k=1}^{26} k^{-3}$ is the truncated Ramanujan zeta regularisation. Inter-level tunnelling rates are computed via the WKB approximation, and phonon-stabilised equilibrium frequencies are derived at each level.

1. Introduction

The cosmological constant problem — the 10^{120} discrepancy between quantum field theory predictions and the observed vacuum energy density $\rho_{\text{vac},0} \approx 5.96 \times 10^{-27} \text{ kg/m}^3$ — remains one of the deepest puzzles in theoretical physics. The UQFF framework approaches this through 26-dimensional compressed gravity, where each dimension contributes a distinct vacuum energy scale.

Previous work (PAPER_1106, PAPER_1107) established the 26D geometric folding operator and the VDS/DVP/BH unified number system. Here we derive the explicit vacuum density at each of the 26 levels, providing a ladder of energy scales that connects quantum gravity to cosmological observations.

2. Ramanujan Zeta Regularisation

The regularisation factor employs the truncated zeta function:

$$S_{26}^{(3)} = \sum_{k=1}^{26} k^{-3} \approx 1.2019286841$$

This converges rapidly to $\zeta(3) \approx 1.2020569$ (Apéry's constant), with the truncation at $k = 26$ reflecting the dimensionality of the UQFF framework.

3. Vacuum Density Hierarchy

3.1 SCm-Corrected Base Density

$$\rho_{\text{SCm}} = \rho_{\text{vac},0} \cdot H_{\text{SCm}} \cdot (1 - \kappa)$$

With calibrated constants $H_{\text{SCm}} \approx 0.99$ and $\kappa = 0.0005 \text{ day}^{-1}$:

$$\rho_{\text{SCm}} \approx 5.894 \times 10^{-27} \text{ kg/m}^3$$

3.2 Level-Dependent Scaling

The dimensional scaling factor at level n :

$$\delta_n = (2\pi)^{n/6}$$

This produces an exponential hierarchy spanning: - Level 1: $\delta_1 = (2\pi)^{1/6} \approx 1.349$ - Level 13: $\delta_{13} = (2\pi)^{13/6} \approx 34.72$ - Level 26: $\delta_{26} = (2\pi)^{26/6} \approx 1,206$

3.3 Complete Ladder

$$\rho_{\text{vac}}^{(n)} = \rho_{\text{SCm}} \cdot S_{26}^{(3)} \cdot (2\pi)^{n/6}, \quad n = 1 \dots 26$$

The cumulative vacuum energy:

$$\rho_{\text{cum}} = \sum_{n=1}^{26} \rho_{\text{vac}}^{(n)} = \rho_{\text{SCm}} \cdot S_{26}^{(3)} \cdot \sum_{n=1}^{26} (2\pi)^{n/6}$$

4. Inter-Level Tunnelling

4.1 WKB Approximation

The tunnelling rate between adjacent levels n and $n + 1$:

$$\Gamma_{\text{WKB}}(n \rightarrow n + 1) = \hbar^{-1} \exp \left(-\frac{\Delta\rho_{n,n+1} \cdot \hbar}{c^2 \cdot \hbar} \right)$$

where $\Delta\rho_{n,n+1} = |\rho_{\text{vac}}^{(n+1)} - \rho_{\text{vac}}^{(n)}|$.

4.2 Decay Cascade

For levels where $\Gamma_{\text{WKB}} > 0$, the vacuum state undergoes a cascade:

$$\frac{d\rho^{(n)}}{dt} = -\Gamma_{\text{WKB}}(n \rightarrow n + 1) \cdot \rho^{(n)} + \Gamma_{\text{WKB}}(n - 1 \rightarrow n) \cdot \rho^{(n-1)}$$

5. Phonon-Stabilised Equilibria

At each vacuum level, the phonon equilibrium frequency:

$$\omega_{\text{eq}}^{(n)} = \frac{\sqrt{\rho_{\text{vac}}^{(n)} \cdot G}}{\hbar}$$

This connects the vacuum density hierarchy to observable phonon spectra in condensed matter analogue systems.

6. Dark Energy Pressure

The inter-level dark energy pressure:

$$P_n = -\rho_{\text{vac}}^{(n)} \cdot c^2$$

The total dark energy density from all 26 levels provides a prediction that can be compared against the observed value $\rho_\Lambda \approx 5.96 \times 10^{-27} \text{ kg/m}^3$.

7. Conclusion

The 26-level vacuum density ladder provides a structured framework for understanding the cosmological constant hierarchy through dimensional decomposition. The Ramanujan zeta regularisation anchors the scaling, while WKB tunnelling and phonon equilibria connect the vacuum structure to observable phenomena.

References

- PAPER_970: Ramanujan S_{26} Application in UQFF Inflation
- PAPER_1106: SCm String Theory 26D Action
- PAPER_1107: UQFF 26D Geometric Folding Operator
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- Weinberg, S. (1989). The cosmological constant problem. Rev. Mod. Phys. 61, 1-23